

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 37.3

$$\int \frac{dx}{x^2 + 2x + 5} = \int \frac{dx}{(x+1)^2 + 4}$$

$$t = x + 1 \ (\Rightarrow \ dt = dx)$$

$$\int \frac{dt}{t^2 + 4}$$

$$= \frac{1}{2} \operatorname{Bgsin} \left(\frac{t}{2} \right) + C$$

$$x + 1 = t$$

$$= \frac{1}{2} \operatorname{Bgsin} \left(\frac{x+1}{2} \right) + C$$

References